Research Statement

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I specialize in algebraic geometry, more specifically Hodge theory and its interactions with moduli spaces and topology of complex varieties. Classical Hodge theory studies complex algebraic varieties by imposing structures on their singular cohomology: a Hodge decomposition in the smooth projective case, and mixed Hodge structures in general. Nonabelian Hodge theory asks how algebraic geometry constrains fundamental groups of these varieties, by investigating monodromy representations and the geometry of their moduli spaces.

In the projective case, its central correspondence relates flat bundles, hence monodromy representations, to Higgs bundles through harmonic theory, bringing algebraic tools such as stability to the study of local systems. The quasiprojective case, on the other hand, is still a work in progress. My research fills in some gaps for the quasiprojective theory. My other projects involve applying Hodge-theoretic methods to cohomology jump loci and Euler characteristics.

Nonabelian Hodge theory for quasiprojective varieties (Section 1). I prove that the nilpotent-residue correspondence is a homeomorphism of moduli spaces in every dimension [Tra26a]. On curves, I extend the correspondence from polystable objects to an exact equivalence between logarithmic flat bundles and semistable logarithmic Higgs bundles, including nonsplit extensions [Tra26c].

Rank-two cohomology jump loci (Section 2). In [Tra26b], I construct Hodge-theoretic structures on jump loci in every rank, on their images in the GIT quotient, and their normalizations. In rank two, I give an explicit classification of irreducible components up to normalization, in terms of algebraic tori, finite quotients, and character varieties of Deligne-Mumford curves.

Positivity of perverse Euler characteristics (Section 3). We prove nonnegativity of perverse Euler characteristics on finite-volume Hermitian locally symmetric spaces of noncompact type, including all Shimura varieties, and strict positivity for full support [AT26]. Applications include the moduli stacks of smooth cubic threefolds and fourfolds.

1. Nonabelian Hodge theory for quasiprojective varieties

For a smooth projective variety X, Simpson's correspondence [Sim92] identifies flat bundles with semistable Higgs bundles with vanishing Chern classes. On semisimple objects, this was achieved by equipping C^∞-flat bundles with harmonic metrics, and induces a homeomorphism between the good moduli spaces $M_{\mathrm{DR}}(X,r)$ and $M_{\mathrm{Dol}}(X,r)$. The Riemann-Hilbert correspondence identifies $M_{\mathrm{DR}}(X,r)$ analytically with the character variety

$$M_{\mathrm{B}}(X,r) = \mathrm{Hom}(\pi_1(X), \mathrm{GL}_r(\mathbb{C}))/\!/\,\mathrm{GL}_r,$$

whose points represent semisimple local systems up to isomorphism. Scaling Higgs fields gives a $\mathbb{C}^*$-action on $M_{\mathrm{Dol}}(X,r)$ that transports to $M_{\mathrm{B}}(X,r)$ via the homeomorphism; rigid local systems are fixed and therefore underlie variations of Hodge structure. Simpson used the resulting restrictions on monodromy to exclude certain lattices, such as $\mathrm{SL}_r(\mathbb{Z})$ for $r \geq 3$, as projective fundamental groups. The full correspondence, including extensions, also enters Cao's proof that the Albanese map of a projective manifold with nef anticanonical bundle is locally trivial [Cao19].

For $X = \bar{X} \setminus D$, with $\bar{X}$ smooth projective and D a simple normal-crossings divisor, a local system with unipotent boundary monodromy has a canonical logarithmic extension with nilpotent residues. Its Higgs counterpart should have nilpotent residues and vanishing Chern classes. Controlling singular behavior along D makes harmonic theory more delicate. Simpson's and Mochizuki's tame harmonic theory supplies the correspondence on polystable objects [Sim90; Moc09]; my work addresses extensions, as well as the induced correspondence map on moduli spaces.

One setting where such comparison results matter is the Shafarevich program: the classical conjecture asks whether universal covers of smooth projective varieties are holomorphically convex.

Eyssidieux-Katzarkov-Pantev-Ramachandran proved the conjecture for smooth projective varieties with linear fundamental groups [Eys+12]. Bakker-Brunebarbe-Tsimerman prove a nonproper linear version and construct algebraic Shafarevich morphisms, which contract precisely the subvarieties on which a chosen local system has finite monodromy [BBT24]. One input into their result is a homeomorphism, given by the nilpotent-residue correspondence, of moduli spaces for curves. I generalize this to higher dimensions.

Theorem A (Tran [Tra26a]). *For a smooth projective log pair $(\bar{X}, D)$, the nonabelian Hodge correspondence induces a homeomorphism in the complex analytic topology*

$$M_{\mathrm{Dol}}^{\mathrm{nilp}}(\bar{X}, D, r) \xrightarrow{\sim} M_{\mathrm{DR}}^{\mathrm{nilp}}(\bar{X}, D, r).$$

The Dolbeault space parametrizes polystable logarithmic Higgs bundles with nilpotent residues and vanishing Chern classes; the de Rham space parametrizes semisimple logarithmic flat bundles with nilpotent residues.

The restriction to nilpotent residues is not artificial: the most general nonabelian Hodge correspondence involves parabolic structures. These introduce non-holomorphic parameters, and to get algebraic moduli varieties, we must fix the eigenvalues of local monodromy.

The importance of the topology of this correspondence is further illustrated by the $P = W$ conjecture of de Cataldo-Hausel-Migliorini [CHM12], proved for GL_n in coprime rank and degree on curves of genus at least two by Maulik-Shen [MS24]. Under nonabelian Hodge theory, the perverse filtration from the Hitchin map agrees, after reindexing, with the weight filtration on character variety's cohomology. This explains an unexpected Lefschetz phenomenon for cohomology of character varieties [HR08]. My homeomorphism provides the groundwork to ask analogous cohomological questions in the higher-dimensional nilpotent-residue setting.

The good moduli spaces retain only semisimplifications, while extension data carry further information about the fundamental group. On curves, I recover a correspondence for extensions.

Theorem B (Tran [Tra26c]). *For a smooth projective curve $\bar{C}$ and D a finite set of reduced points, there is an exact $\mathbb{C}$-linear equivalence of abelian categories*

$$\mathsf{Conn}_{\mathrm{nilp}}(\bar{C}, D) \xrightarrow{\sim} \mathsf{Higgs}_{\mathrm{nilp}}^{\mathrm{ss},\deg 0}(\bar{C}, D)$$

from logarithmic flat bundles with nilpotent residues to semistable degree-zero logarithmic Higgs bundles with nilpotent residues. It extends the tame correspondence on polystable objects.

Nonsplit extensions need not carry harmonic metrics, and projective arguments using Kähler formality and the $\partial\bar{\partial}$-lemma [Sim92] do not directly give the needed logarithmic comparison. Instead, my construction treats extensions as deformations of their semisimplification with respect to the canonical socle filtration. The relevant deformation problems are identified, which allows transporting deformations and then algebraizing. A higher-dimensional comparison, potentially using multi-V-filtrations, should generalize the same construction.

2. Rank-two cohomology jump loci

Let X be a smooth, connected, and projective variety. On the representation space $R_{\mathrm{B}}(X, x, r) = \mathrm{Hom}(\pi_1(X, x), \mathrm{GL}_r(\mathbb{C}))$, the jump loci

$$\Sigma_k^i(X, x, r) = \{V : \dim H^i(X, V) \geq k\}$$

record where cohomology increases in dimension. Write $\overline{\Sigma}_k^i(X, r)$ for their reduced images in $M_{\mathrm{B}}(X, r)$. In rank one, Green-Lazarsfeld's linearity results for cohomology support loci of line bundles [GL91] gives a structure theorem: rank one jump components are translates of subtori. Simpson [Sim93] later proved that these are torsion-translated, and Budur-Wang extended this to

smooth quasiprojective varieties [BW15b]. Such results give effective restrictions on fundamental groups of algebraic varieties [DPS09].

In higher rank, the local picture has been studied extensively through deformation theory and formality [GM88; Sim92]. Dimca-Papadima [DP14] model nonabelian jump locus germs near the trivial representation using commutative differential graded algebras; Budur-Wang [BW15a] develop deformation theory with cohomological constraints using differential graded Lie algebra pairs. Papadima-Suciu [PS19; PS20] obtain rank two decomposition and pullback results of germs. However, the global geometry of higher-rank jump loci remains largely unknown. My paper [Tra26b] gives both Hodge structures in every rank and a global structure theorem in rank two.

Theorem C (Tran [Tra26b], Hodge structures on jump loci)**.** *For every rank r, the jump locus $\Sigma^i_k(X,x,r)$ is a Hodge subvariety. Write Σ for its image in $M_{\mathrm{B}}(X,r)$. If $V\in\Sigma$ underlies a complex variation of Hodge structure, then the completed local rings*

$$\widehat{\mathcal{O}}_{\Sigma,V},\qquad \widehat{\mathcal{O}}_{\Sigma,V},\qquad \widehat{\mathcal{O}}_{Z^\nu,z},$$

carry compatible pro-mixed Hodge algebra structures, where Z is any irreducible component of Σ containing V, Z^ν is its normalization, and $z\in Z^\nu$ lies over V.

These structures constrain the local equations of jump loci in every infinitesimal neighborhood. Their compatibility with the GIT quotient and normalization makes the Hodge-theoretic information available on the spaces appearing in the global classification below. More generally, I obtain mixed twistor structures on formal deformations at every semisimple $V\in\Sigma$. The following classification relies on Corlette-Simpson's theorem on SL_2-representations [CS08].

Theorem D (Tran [Tra26b])**.** *Let Z be an irreducible component of $\overline{\Sigma}^i_k(X,2)$.*

(1) In the reducible case, $Z\simeq \mathrm{Sym}^2(N)$ for a translated torus N, or Z^ν is a product of two translated tori. In the irreducible dihedral case, Z^ν is a torus or Z is a quotient of a translated torus by an involution. In the projectively finite and projectively dense rigid cases, Z^ν is a torus.

(2) In the projectively dense nonrigid case, either Z^ν is a torus or

$$Z^\nu\simeq (Q^\nu\times T)/H,$$

where T is an algebraic torus, Q is a fixed-determinant, fixed-inertia character variety component of a proper hyperbolic Deligne-Mumford curve, and H is finite and acts freely.

Here Z^ν and Q^ν denote normalizations.

This fits the broader program of characterizing "motivic subvarieties" of character varieties surveyed by Litt [Lit24, §§5.5, 6.4.2]. Jump loci provide a concrete class in which to test these structural expectations; their classification may also yield fundamental group obstructions beyond rank one.

Absolute $\overline{\mathbb{Q}}$-closed loci are Hodge subvarieties in rank one: they are finite unions of torsion translates of triple tori [Sim93; BW20]. Whether absolute loci are Hodge in higher rank is a natural question; my results provide positive evidence since reduced jump loci and their irreducible components are both absolute and Hodge. The weaker question of whether absolute $\overline{\mathbb{Q}}$-constructible subsets are $\mathbb{C}^*$-stable was posed by Budur-Wang [BW20, Remark 7.4.2], with special case results due to Lerer [Ler22] and Budur-Lerer-Wang [BLW23].

3. Positivity of perverse Euler characteristics

The Chern-Hopf-Thurston sign conjecture predicts $(-1)^n\chi(X)\geq 0$ for a closed aspherical real $2n$-manifold. On a smooth complex n-fold, $\mathbb{Q}_X[n]$ is perverse, so nonnegativity of $\chi(X,\mathcal{P})$ for every perverse sheaf $\mathcal{P}$ strengthens the corresponding sign inequality. Liu-Maxim-Wang propose

this perverse refinement for aspherical smooth projective varieties [LMW21, Conjecture 6.2], and Arapura-Wang formulate it for compact aspherical Kähler manifolds [AW25, Conjecture 1.2].

A foundational example is Franecki-Kapranov's nonnegativity theorem for perverse sheaves on semiabelian varieties [FK00]. On an abelian variety A, generic vanishing gives another explanation: for a perverse sheaf $\mathcal{P}$ and a general rank-one local system L, one has $H^j(A, \mathcal{P} \otimes L) = 0$ for $j \neq 0$, hence

$$\chi(A, \mathcal{P}) = \chi(A, \mathcal{P} \otimes L) = \dim H^0(A, \mathcal{P} \otimes L) \geq 0.$$

Bhatt-Schnell-Scholze prove generic vanishing topologically and bound the codimensions of its exceptional loci [BSS18]. This makes a concrete connection with Section 2: restrictions on where twisted cohomology survives can force Euler-characteristic inequalities.

Arapura-Wang proved positivity in several projective settings, including varieties carrying a faithful rigid local system [AW25]. Arapura-Patel established quasiprojective results for moduli stacks of curves and abelian varieties and for Shimura varieties of Hodge type [AP26]. Their prediction for all Shimura varieties follows from joint work with Mert Akdenizli.

Theorem E (Akdenizli-Tran [AT26])**.** *Let $\mathcal{D}$ be a Hermitian symmetric domain of noncompact type, and $\Gamma \subset \mathrm{Aut}(\mathcal{D})^\circ$ a lattice. Denote $\mathcal{X} = [\mathcal{D}/\Gamma]$. For every rational perverse sheaf $\mathcal{P}$,*

$$\chi_{\mathrm{orb}}(\mathcal{X}, \mathcal{P}) \geq 0$$

with strict inequality for full support. For torsion-free Γ, this is the ordinary Euler characteristic of the smooth variety $\Gamma\backslash\mathcal{D}$.

The applications extend beyond locally symmetric spaces. Finite maps and affine locally closed immersions preserve perversity under derived pushforward, with $\chi(Y, \mathcal{P}) = \chi(X, Rf_*\mathcal{P})$. Nonnegativity therefore passes to closed subvarieties, complements of effective Cartier divisors, and their finite covers. Together with period descriptions, this gives the applications to moduli stacks of smooth cubic threefolds and fourfolds.

Taking the intersection complex of a closed irreducible subvariety Z also gives $(-1)^{\dim Z}\chi_{\mathrm{IH}}(Z) \geq 0$. Thus the theorem controls intersection cohomology of singular subvarieties as well as ordinary Euler characteristics, and establishes a noncompact counterpart of the perverse Singer-Hopf prediction.

There is also a hyperbolicity connection: nonnegativity for all perverse sheaves rules out every nonconstant $f : \mathbb{P}^1 \to X$, since f would be finite and $\chi(X, f_*\mathbb{Q}_{\mathbb{P}^1}[1]) = -2$. Arapura's recent conjecture [Ara25, Conjecture 3.3] links Kollár hyperbolicity - nontrivial étale monodromy along every nonconstant map from a smooth projective curve - to asymptotic cohomology vanishing in towers of étale covers. This predicts perverse Euler nonnegativity for sheaves underlying mixed Hodge modules on normal projective varieties with that property.

4. Future projects

4.1. **Logarithmic de Rham-Dolbeault cohomology comparison.** For a harmonic bundle on a smooth projective variety, Simpson identifies the hypercohomology of its de Rham and Dolbeault complexes [Sim92]. On a quasiprojective variety, logarithmic comparison is known for bundles underlying admissible variations of mixed twistor structures. The extension correspondence raises the question of comparison for the resulting nonpolystable matchings.

Using ideas from Katzarkov-Pantev-Toën [KPT09], I have obtained a comparison for $(\bar{X}, D)$ admitting iterated fibrations by smooth projective log curves. Outside the logarithmic setting, reductions using towers of affine curves are available for smooth quasiprojective varieties. The difficulty here is to preserve the chosen compactification and boundary. I plan to investigate controlled modifications of log pairs that admit such fibrations.

4.2. **Parabolic nonabelian Hodge correspondence.** Local systems of geometric origin generally have quasi-unipotent boundary monodromy. Their Dolbeault counterparts are parabolic Higgs bundles with rational weights and vanishing parabolic Chern classes; Mochizuki's tame correspondence already treats polystable objects in this setting [Moc09]. I plan to extend Theorems A and B to parabolic bundles.

Finite ramified covers can make the local monodromy unipotent, but moduli applications require a uniform construction for fixed boundary data. Root stacks provide this framework through the correspondence between parabolic bundles and bundles on a root stack [Bor07; BV12]. I will develop the necessary harmonic and twistor theory on these nice stacks. Simpson's work on local systems on algebraic V-manifolds [Sim11] gives a precedent.

4.3. **Rank-three cohomology jump loci.** Rank three is the natural next step after Theorem D, but there is no comparable classification of nonrigid representations. In the rigid case, Langer-Simpson prove that every rigid, integral, irreducible SL_3-representation of a smooth projective fundamental group is of geometric origin [LS18], providing evidence for Simpson's motivicity conjecture. The jump loci problem asks for structure on positive-dimensional families as well. My Hodge subvariety results suggest using the Shafarevich morphisms of [BBT24]. For a jump component Z, a plausible strategy is to work with a morphism $\mathrm{sh}_Z : X \to \mathcal{Y}_Z$ to a Deligne-Mumford stack through which all local systems in Z descend. Writing $V = \mathrm{sh}_Z^* W$, the projection formula gives

$$R\Gamma(X, V) \simeq R\Gamma\big(\mathcal{Y}_Z, W \otimes R(\mathrm{sh}_Z)_* \mathbb{C}_X\big).$$

This turns the question into one about twisted jump loci on $\mathcal{Y}_Z$. I want to identify conditions on X that make sh_Z tractable, and investigate whether $\dim \mathcal{Y}_Z$ can be bounded for rank-three components with positive-dimensional image in the PGL_3-character variety. The rank-two proof analyzes twisted jump loci on Deligne-Mumford curves; developing a corresponding analysis for Deligne-Mumford surfaces is a concrete next problem.

4.4. **Hodge-theoretic and KSBA compactifications.** Torelli theorems relate moduli spaces to images of period maps, as is the case for abelian varieties and K3 surfaces, and generically for most hypersurfaces. When the period domain is Hermitian symmetric, the Satake-Baily-Borel compactification [BB66] provides a Hodge-theoretic boundary. Recently, Bakker-Filipazzi-Mauri-Tsimerman [Bak+25] constructed compactifications of period images much more generally. These boundaries record degenerations of Hodge structures, but need not specify a limiting algebraic variety. I want to compare them with modular compactifications; one prominent class is KSBA compactifications, whose points represent stable varieties or pairs.

There are compelling precedents. The main component of Alexeev's compactification by semi-abelic pairs is a toroidal compactification of the moduli of principally polarized abelian varieties [Ale02]. Alexeev-Engel [AE23] identify certain KSBA compactifications of polarized K3 surfaces as semitoroidal. These modular compactifications retain geometric boundary data contracted by the Satake-Baily-Borel compactifications.

I plan to start with canonically polarized hypersurfaces $X_d \subset \mathbb{P}^{n+1}$ with $d \geq n+3$, beginning with quintic surfaces in $\mathbb{P}^3$. Gallardo describes their GIT compactification and boundary, including stability of normal quintics having at worst isolated double points or minimal elliptic singularities [Gal19]. Rana describes a KSBA boundary divisor associated with stable numerical quintic surfaces whose only non-Du Val singularity is a Wahl singularity [Ran17]. These results provide explicit degenerations and stable models, which can be used to compare their GIT limits, KSBA replacements, and limiting Hodge data.

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